

Accelerate Science Now Response to Office of Science and Technology Policy Request for Information on Artificial Intelligence (AI) Action Plan

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I. Introduction

As AI continues to drive innovation across scientific research, national security, and industrial applications, the United States must take decisive action to maintain its leadership in AI development and deployment. In 2021, Congress established the National AI Initiative to further coordinate and enhance Federal actions to ensure continued U.S. leadership in AI research and development, leading the world in the development and use of trustworthy AI systems in the public and private sectors.

The Initiative aims to prepare the present and future U.S. workforce for the integration of AI systems across all sectors of the economy and society, and coordinates ongoing AI research, development, and demonstration activities among the civilian agencies, Department of Defense, and the Intelligence Community to ensure that each informs the work of the others. A comprehensive national strategy should build on the work started by the National AI Initiative and prioritize investments in cutting-edge research, energy infrastructure and public-private collaboration to ensure the responsible and effective advancement of AI technologies.

The following recommendations outline key initiatives to empower federal agencies, enhance scientific research capabilities, and address the critical challenges associated with AI adoption, governance, and security. By implementing these strategies, the U.S. can make groundbreaking scientific advancements, strengthen economic competitiveness, and reinforce national security while ensuring that AI-driven advancements remain aligned with ethical and societal goals.

Accelerate Science Now is a non-partisan coalition of leaders in industry, academia, civil society, and the research community, charged with igniting a new era of rapid scientific discovery and delivering the benefits to the American people. Accelerate Science Now members include: *Amazon Web Services; Anthropic; Astera Institute; Black Tech Street; Carnegie Mellon University; Center for Data Innovation; Cohere; Computing Research Association; Emerald Cloud Lab; Energy Sciences Coalition; Federation of American Scientists; Foundation for*

American Innovation; FutureHouse; Ginkgo Bioworks; Institute of Electrical and Electronics Engineers; Institute for Progress; Institute for AI Policy and Strategy; Information Technology Industry Council; Lehigh University; Meta; National Applied AI Consortium; New Mexico Artificial Intelligence Consortium - Academia; NobleReach Foundation; OpenMined; Renaissance Philanthropy; Samsung; SeedAI; UbiQD; UC Berkeley; UC Irvine, University of Florida; University of Wisconsin-Madison; and the Wilson Center Science and Technology Innovation Program.

Accelerate Science Now is led by SeedAI, a non-profit, nonpartisan organization working at the forefront of artificial intelligence policy and governance.

These recommendations do not necessarily represent or reflect the official positions of all coalition members. This document should be understood as a collaborative effort to advance shared objectives, while acknowledging the diversity of viewpoints within our coalition.

II. Recommendations

Emerging Ideas

- **Empowering OSTP to Lead on AI and Scientific Research Priorities**

To advance the United States' position as a global leader in AI and scientific innovation, the Office of Science and Technology Policy (OSTP) should be empowered to take a more active role in shaping national research priorities.¹

Key Considerations:

- OSTP should develop a Scientific Grand Challenge Agenda.
- OSTP should coordinate a pilot program for leveraging expert prediction markets to inform AI R&D activities.

OSTP has a substantial amount of scientific and technological expertise at its disposal, but its ability to directly shape the federal government's R&D priorities is limited. For example, the Director of OSTP was elevated to a cabinet-level position in 2022, but without commensurate cabinet-level authority.

¹ The National AI Initiative Act, passed as part of the National Defense Authorization Act of 2020 in the first Trump Administration, provided new but limited authority to the Office of Science and Technology Policy with regards to AI policy. For example, it gave it the authority to name the Director of the National AI Initiative Office and designate and co-chair an interagency committee to coordinate federal programs in support of the office. These authorities expire 10 years after enactment. National AI Initiative Act. 15 U.S.C. § 9412-9413 (2021).

Through agencies like the Department of Energy (DOE), the National Institutes of Health (NIH), and the National Science Foundation (NSF), the Federal Government already has effective mechanisms and well-established expertise for coordinating and leading research and development itself and in partnership with academia and the private sector. Thus it is critical that empowering OSTP to exhibit greater leadership in this space is additive, rather than coming at the expense of existing federal R&D leadership efforts, or is duplicative of their work.

To accomplish this, OSTP should be granted greater authority to coordinate ambitious interagency R&D agendas and experiment with new methods for advancing national strategic objectives and scientific and technological competitiveness.

More concretely, OSTP should be empowered to set ambitious federal R&D targets by developing a Scientific Grand Challenge Agenda and coordinating a pilot program for leveraging expert prediction markets to inform AI R&D activities.

- **Scientific Grand Challenge Agenda**

OSTP should develop a high level Scientific Grand Challenge Agenda to serve as a roadmap for public and private R&D efforts in science and technology.

Key Considerations:

- The Agenda should be a persistent mechanism to chart the course for national scientific and technical leadership.
- The Agenda should identify major scientific and technological breakthroughs that would generate enormous economic and social value.
- Federal research agencies and Congress should leverage the Agenda to make more informed decisions about how to best support national R&D priorities.
- OSTP should explore how to leverage the Agenda to inform ambitious, multi-agency funding efforts coupled with non-governmental collaboration.

Many federal agencies have pursued grand challenges as a mechanism for spurring progress towards high value scientific breakthroughs, and they are a critical tool for signaling to academia and the private sector about national research priorities. However, these efforts are typically siloed within individual agencies, or are one-off efforts, limiting their utility as a tool for shaping long-term, ambitious scientific R&D efforts.

The Scientific Grand Challenge Agenda should be a persistent mechanism to chart the course for national scientific and technical leadership. It should be developed in consultation with the Presidential Council of Advisors on Science and Technology (PCAST), leadership of federal research agencies, and input from the public through public requests for comment. OSTP should maintain a regular and iterative process for

updating the Agenda based on changes in the scientific and technical landscape and national priorities.

The list of scientific grand challenges identified in the Agenda should be major scientific and technological breakthroughs that would generate enormous economic and social value. They should place a particular emphasis on the increasing capabilities of AI to augment and automate scientific research, as well as the new opportunity space for scientific exploration enabled by increased AI capabilities.

Federal research agencies should utilize this Scientific Grand Challenge Agenda to inform and augment, but not supplant, their own R&D priorities. And Congress should leverage the Agenda to make more informed decisions about how to best support national scientific and technological priorities.

OSTP should also explore how to leverage the Agenda to inform ambitious, multi-agency funding efforts that incentivize research teams from academia and the private sector to collaborate and make shared, iterative progress towards solving these challenges. This could help bypass the “lone-ranger” paradigm for scientific research and more effectively ensure that federal R&D efforts produce high-quality science and deliver for the American people.²

The Apollo Program, the Human Genome Project, and the DARPA Grand Challenge are all excellent examples of ambitious grand challenges that successfully mobilized coordinated, multi-stakeholder, long-term efforts to drive groundbreaking scientific and technological progress.^{3,4,5} The federal government should make systematically identifying and pursuing groundbreaking ideas like these a consistent, whole-of-government priority rather than one-off projects.

- **Pilot Program for Expert Prediction Markets in AI R&D**

To enhance the effectiveness of AI research funding, OSTP, in coordination with DOE, should establish a pilot program exploring the utility of expert prediction markets to inform AI R&D funding decisions.

Key Considerations:

² Mitchell, Tom M. (2024) *How Can AI Accelerate Science, and How Can Our Government Help?* Carnegie Mellon University [online] <https://tinyurl.com/2s3ctsf8>

³ Mann, A. (2020, June 25). *The apollo program: How NASA sent astronauts to the Moon*. Space.com. <https://www.space.com/apollo-program-overview.html>

⁴ National Institute of Health. (2024, June 14). *Human genome project fact sheet*. Genome.gov. <https://www.genome.gov/about-genomics/educational-resources/fact-sheets/human-genome-project>

⁵ *The DARPA Grand Challenge: Ten Years Later*. DARPA. (2014, March 13). <https://www.darpa.mil/news/2014/grand-challenge-ten-years-later>

- OSTP should experiment and iterate on what an expert prediction market looks like.
- OSTP should have a strong bias for including non-government experts open to supporting nontraditional or unexpected approaches.
- OSTP should develop a rigorous evaluative framework to compare the outcomes of prediction market-funded projects with those selected through traditional means.
- OSTP should identify criteria that determine whether particular scientific research priorities are better suited for this expert prediction market approach compared to traditional methods for making research funding decisions.

Research demonstrates the utility of prediction markets across disciplines, including economics, public health, and weather forecasting.⁶ The goals of this pilot should be to 1) identify new methods for evaluating and pursuing promising AI R&D objectives and 2) evaluate the potential for the use of expert prediction markets to shape federal R&D priorities more broadly.

Existing mechanisms for evaluating and reviewing research funding proposals don't always advance high-risk, high-reward research with the highest impact. Reviews can also be biased toward incremental, conservative approaches with more guaranteed rates of success rather than revolutionary science and engineering solutions.⁷ Compounding this problem, over more than a decade the rate of success across federal science agencies for new applications has been very low, especially at NIH and NSF.⁸ For example, in 2023, NIH received 51,883 grant applications and awarded 11,502 grants—a success rate of 21%. The NIH has experienced roughly 20% success rates for twenty years and hasn't had a success rate of 30% or over since 2003.^{9,10}

OSTP should work with DOE to launch a notice of funding opportunity around a specific technical challenge in AI R&D and DOE should utilize an expert prediction market to augment how it evaluates proposals and guide funding decisions. DOE's evaluation criteria should broaden beyond scientific and technical merit and include potential impact, speed, market potential, and other factors.

⁶ Kenneth J. Arrow *et al*, The Promise of Prediction Markets. *Science* 320, 877-878 (2008). DOI <https://doi.org/10.1126/science.1157679>.

⁷ Axel Philipps, Research funding randomly allocated? A survey of scientists' views on peer review and lottery, *Science and Public Policy*, Volume 49, Issue 3, June 2022, Pages 365–377, <https://doi.org/10.1093/scipol/scab084>

⁸ Düzgüneş N. 'Science by consensus' impedes scientific creativity and progress: A simple alternative to funding biomedical research. *F1000Res*. 2024;11:961. Published 2024 Feb 21. doi:10.12688/f1000research.124082.3

⁹ NIH IMPAC, Success Rate File - Data produced by Statistical Analysis and Reporting

¹⁰ NSF received 40,453 proposals in 2024 and accepted 10,592 for a success rate of 26%. <https://dellweb.bfa.nsf.gov/awdfr3/default.asp>

OSTP should be willing to experiment and iterate on what this expert prediction market looks like, but it should include a mix of internal and external subject matter experts voting on which proposals they think are most likely to deliver positive results. To meaningfully differ from the traditional approach of internal expert review, OSTP should have a strong bias for including non-government experts that might be willing to support nontraditional or unexpected approaches. OSTP should also experiment with an incentive system that rewards participating experts for correct predictions – those that successfully deliver positive research results – while also carefully protecting against potential conflicts of interest.

OSTP should develop a rigorous evaluative framework to compare the outcomes of prediction market-funded projects with those selected through traditional means, carefully considering factors like cost, speed, quality, and innovation. And importantly, OSTP should identify criteria that determine whether particular scientific research priorities are better suited for this expert prediction market approach compared to traditional methods for making research funding decisions. Based on lessons learned from this pilot, if results are positive, OSTP should determine how to scale this approach across the federal research ecosystem and augment existing approaches to funding scientific research.

Public-Private Collaboration and Government AI Adoption

- **Government AI Adoption**

The adoption of AI within government agencies presents an unparalleled opportunity to enhance scientific research, improve decision-making, and optimize public services. The Administration, through OSTP and in consultation with PCAST, should encourage the use of open AI models across the executive branch, allowing for increased transparency and collaboration among researchers and policymakers. By integrating AI-driven insights, government agencies can improve policy formation, optimize resource allocation, and foster public trust in AI-driven governance.

- **Empower Scientific Workforce**

To maintain the U.S. position as a global leader in AI and scientific research, the federal government should expand AI literacy across the scientific workforce. The Administration should establish training programs and funding initiatives aimed at small and medium-sized businesses engaged in R&D. These efforts should leverage existing programs and relationships through the NSF, DOE, Department of Agriculture (USDA), and Small Business Administration (SBA), as well as any other executive agencies identified by OSTP in consultation with PCAST. AI-enhanced tools can improve productivity, facilitate interdisciplinary research, and unlock new scientific frontiers. A concerted effort to integrate AI training into federal workforce development programs

and academic institutions will ensure that researchers have the skills necessary to leverage AI for discovery and technological advancement.

- **Establish NIST Foundation**

The Administration should establish a dedicated foundation at the National Institute of Standards and Technology (NIST) to significantly enhance AI standards development and public-private collaboration. This foundation should support:

- Development of AI safety and ethical standards
- Expansion of partnerships between research institutions, industry, and government
- Research grants and pilot projects focused on AI applications in scientific discovery and industrial innovation
- Workforce development initiatives aligned with emerging AI technologies

- **Empower and expand the National AI Research Resource (NAIRR)**

The Administration should fully support the National AI Research Resource (NAIRR). Conceived during the first Trump Administration, the NAIRR is critical in advancing U.S. competitiveness in AI and the Administration should ensure it has the resources necessary for continued success. Sustained investments in AI R&D have enabled the United States to be a longstanding global leader in the field of AI, from the foundations of the field to the present day. Realizing the benefits of AI for the United States will rely on the ability of all U.S. researchers to access the necessary resources, especially researchers with limited access or who have been historically excluded from AI and related fields and industries. Engaging the full entire pool of U.S. talent will bring important perspectives, research capacity, and inspiring use cases.

The NAIRR would provide a widely-accessible, national cyberinfrastructure that empowers a diverse set of users across a range of fields through access to computational, data, and training resources. Leveraging the U.S.'s existing cyberinfrastructure resources, the NAIRR would support cutting-edge explorations in AI R&D and improve the ease of collaboration across disciplines and sectors that address pressing problems with AI. It would create opportunities to train the future AI workforce, support and advance trustworthy and responsible AI, and catalyze development of ideas that can be practically deployed for societal and economic benefits.

The NAIRR would act as a catalyst for collaborative projects among universities, private industry, and government agencies. By offering a centralized platform for data and computing power, the NAIRR can streamline interdisciplinary research, reduce redundancy, and foster a culture of open innovation.

The enhanced capabilities provided by NAIRR can spur the development of new AI-driven applications across critical sectors such as healthcare, cybersecurity, and

transportation, driving job creation and bolstering economic competitiveness on both a national and global scale.

The National Science Foundation established a NAIRR Pilot Program that demonstrated how successful a fully funded NAIRR could be.¹¹ The pilot program has partnered with 14 governmental and over 25 non-governmental organizations and supported over 250 research projects in more than 40 states. Unfortunately, despite bipartisan support, Congress has not authorized or appropriated funding for the NAIRR and the pilot program has less than a year left until it expires.

The Trump Administration can and should continue to champion the NAIRR, regardless of Congressional action. It should build on the success of the pilot to scale up and expand, leveraging existing supporters in industry and the nonprofit sector to recruit additional non-governmental actors, such as nonprofits, foundations, and decentralized networks. Private sector actors are critical for the success of the NAIRR and should be fully leveraged to augment governmental action.

- **Support the Department of Energy’s Frontiers in Artificial Intelligence for Science, Security and Technology (FASST) Initiative**

The Administration should support full funding for the DOE FASST Initiative. This transformative program leverages the Department’s extensive resources and established infrastructure to address critical national priorities. By integrating artificial intelligence with science, security, and technology, FASST is uniquely positioned to deliver advancements that bolster national security, drive scientific discovery, and overcome energy challenges—all while nurturing a skilled, forward-thinking workforce.

The FASST Initiative will advance national security by enhancing research capabilities through integrating AI with existing DOE capabilities and enabling rapid data analysis and predictive modeling. It will provide a technological edge by harnessing AI to strengthen defense and cybersecurity measures and foster partnerships with defense, industry, and academic institutions to address emerging security threats.

A fully funded FASST will support robust training programs, fellowships, and partnerships with leading research institutions, positioning FASST as a magnet for top-tier talent in AI, data science, and engineering. FASST investments in education and workforce training will ensure a pipeline of skilled professionals who can drive the long-term success of national security and energy innovation initiatives.

The integration of AI technologies within DOE’s scientific research programs will dramatically reduce the time required for data processing and simulation, expediting

¹¹ *National Artificial Intelligence Research Resource Pilot*. U.S. National Science Foundation. (n.d.). <https://www.nsf.gov/focus-areas/artificial-intelligence/nairr>

breakthroughs in physics, chemistry, and materials science. AI's application across various scientific domains will enable researchers to tackle complex challenges with enhanced precision and efficiency. FASST can leverage DOE's vast computational and experimental resources to enhance the accuracy and scalability of scientific investigations, paving the way for next-generation discoveries.

Energy and Computing Capacity

Energy and compute infrastructure are critical components for all AI applications, including scientific research, industrial automation, and large-scale data processing. As AI models grow in complexity, their computational requirements increase exponentially, leading to higher energy consumption. As a result, the United States' energy needs to power AI are growing rapidly, with data center energy demand expected by some experts to exceed 100 GW by 2030.¹² Ensuring access to affordable and efficient compute and energy resources is especially important for research institutions, which often struggle to compete with private-sector actors that have access to greater funding and infrastructure. To support the increasing energy demands of AI workloads, the Administration should invest directly, work with states, and alongside private partners, in upgrading our energy infrastructure. Specifically, the Administration should:

- **Modernize the Grid**
Upgrade existing power networks to improve energy efficiency, reliability, and scalability. This can involve replacing aging infrastructure with smart grid technologies, improving energy storage capabilities, and deploying real-time monitoring systems.
- **Integrate Grid-Enhancing Technologies (GETs)**
Leverage advanced power flow control, dynamic line ratings, and other emerging technologies to optimize energy distribution and reduce congestion in transmission lines.
- **Establish Transmission Networks and Dedicated Corridors**
Dedicated transmission networks should be established by the federal government to support the energy-intensive operations of AI data centers and high-performance computing clusters. It should invest in high-voltage direct current (HVDC) transmission lines to efficiently transport electricity over long distances, minimizing energy loss; create dedicated corridors that prioritize power distribution for AI computing hubs; integrating renewable energy sources, advanced grid infrastructure, and redundant power supply pathways to enhance reliability and ensure low-latency access to energy; and, encourage the strategic placement of AI and HPC facilities near renewable energy plants to maximize efficiency and reduce reliance on fossil fuels.

¹² McGeady, C., Majkut, J., Harithas, B., & Smith, K. (2025, March 3). *The electricity supply bottleneck on U.S. AI dominance*. CSIS. <https://www.csis.org/analysis/electricity-supply-bottleneck-us-ai-dominance>

The Administration should also prioritize R&D efforts focused on improving the energy efficiency of AI models and data center infrastructure.

Scientific Applications of AI

AI has made it possible to solve many of the world's hardest challenges. Deploying AI across scientific disciplines will unlock American ingenuity and deliver countless benefits to the American people. To support existing efforts to leverage AI in scientific R&D, the Administration, in consultation with OSTP and PCAST, should:

- **Scientific AI R&D Catalog**

OSTP should create a repository of high-leverage research opportunities and investments that leverage AI to advance science, with input from non-governmental organizations. Whereas the Scientific Grand Challenges Agenda should outline ambitious, moonshot projects, this catalog should identify foundational and high-leverage investments that non-governmental actors could make to advance scientific competitiveness in America. This catalog can serve as a roadmap and facilitate coordinated action from industry, nonprofits, and academia to build on the success of existing projects and augment public sector efforts to support American science.

- **National Lab Testbeds**

Continue to support government investment in AI infrastructure and testbeds at DOE, NIST, and NSF for national security and scientific applications in partnerships with national laboratories to drive cutting-edge research in areas such as nuclear fusion, quantum computing, and medicine.

- **AI-Powered Materials Discovery and Autonomous Laboratories**

Instruct DOE to take a leading role in advancing AI-powered materials discovery to drive breakthroughs in critical material development. This would reinforce U.S. leadership in manufacturing while reducing reliance on foreign supply chains. To accelerate this effort, the U.S. government should establish a National Center for Autonomous Materials Science to leverage AI-driven autonomous laboratories for rapid innovation in materials and manufacturing.¹³ This center should develop standardized, modular lab ecosystems, foster public-private partnerships, and provide funding incentives to overcome adoption barriers. By combining AI-powered discovery with autonomous experimentation, the U.S. can accelerate the identification and synthesis of next-generation materials essential for high-capacity battery storage, next-generation semiconductors, and other strategic technologies. This initiative would revolutionize materials R&D, reduce time-to-market

¹³ Joress, H. , Trautt, Z. , McDannald, A. , DeCost, B. , Kusne, A. and Tavazza, F. (2024), Driving U.S. Innovation in Materials and Manufacturing using AI and Autonomous Labs, Special Publication (NIST SP), National Institute of Standards and Technology, Gaithersburg, MD, [online], <https://doi.org/10.6028/NIST.SP.1320>, https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=958246

for new innovations, and strengthen U.S. global competitiveness in advanced manufacturing.

- **AI Explainability Research**

Invest, through NSF, NIST, and DOE, in advanced AI explainability research to improve reliability and accelerate discoveries in critical fields such as protein folding and disease research. Sound science relies on transparent methods that can be rigorously scrutinized, and explainable AI directly supports that goal. Mechanistic interpretability—one example of explainability research—helps demystify AI decision-making by revealing how individual model components contribute to outcomes, thereby enhancing trustworthiness and robustness in scientific applications. The goal of solving mechanistic interpretability would be a perfect candidate for inclusion in OSTP’s Scientific Grand Challenge Agenda.¹⁴

- **Life Science Data Sets**

Provide funding to support the development of a coordinated initiative for roadmapping, certifying, collecting, and sharing high-quality datasets in the life sciences. This effort should engage nonprofit research organizations and the broader scientific community to identify critical data types for predictive modeling, establish rigorous quality control and open science standards, develop automated data collection methodologies, certify data providers, and subsidize the secure hosting of open datasets.¹⁵ By implementing this strategy, the U.S. can incentivize the creation and sharing of large-scale, high-quality datasets, accelerating scientific discovery and innovation in the life sciences.

III. Conclusion

The United States’ competitiveness in science will be a direct function of its competitiveness in AI. By demonstrating strong leadership in shaping national scientific and technological priorities and strategically providing the resources and infrastructure necessary for effective AI development and deployment, particularly in scientific domains, the Administration’s AI Action Plan can turbocharge American science to rapidly and responsibly deliver the benefits to the American people.

¹⁴ Hammond, S. (2023, May 8). *Opinion | we need a manhattan project for AI safety*. Politico. <https://www.politico.com/news/magazine/2023/05/08/manhattan-project-for-ai-safety-00095779>

¹⁵ DeBenedictis, E., Andrew, B., & Kelly, P. (25AD). *Kickstarting Collaborative, AI-Ready Datasets in the Life Sciences*. Federation of American Scientists. <https://fas.org/publication/collaborative-datasets-life-sciences/>

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